

Use Case 5: Classification of diagnostic standard planes in ultrasound imaging



Spine (sag.): 0.65
Background: 0.165,
Femur 0.01, 3VV 0.01,
Spine (cor.) 0.05,
RVOT 0.05, LVOT 0.05

Problem category: Image Level Classification

Recommended metrics:

- Multi-class Counting Metric: None
- Per-class Counting Metric: [Metric@\(TargetMetric = TargetValue\)](#)
- Multi-threshold Metric: [Area under the Receiver Operating Characteristic Curve \(AUROC\)](#)
- Calibration Metrics: [Class-wise Calibration Error \(CWCE\)](#)

Fingerprint answers:

Fingerprint	Answer
Decision rule (cutoff method) applied to predicted class scores	Target value-based
Penalization of errors: Unequal severity of class confusions	No
Penalization of errors: Unequal interest across classes	Yes
Mismatch between class prevalences and class importance	Yes
Provided class prevalences reflect the population of interest	No
Availability of predicted class scores	Yes
Calibration assessment requested	Yes
Comparative calibration assessment requested	No comparative assessment
Interpretable estimate of calibration error requested	Assessment of calibration error in isolation

Decision between metrics:

- [Calibration Metric]: Expected Calibration Error (ECE) & Root Brier Score (RBS) vs. Classwise Calibration Error (CWCE) vs. Expected Calibration Error Kernel Density Estimate (ECE^{KDE}) & Class-wise Calibration Error (CWCE) & Root Brier Score (RBS)
 - Answer: Classwise Calibration Error (CWCE)